TBL 7.1 STUDY MATERIAL

SUMMARY OF CELL BIOLOGY, CHAPTER 15: Cell division and programmed cell death

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Learning objectives:

• To understand tissue homeostasis as a regulatory mechanism in which an equilibrium between the number of cells that divide and the number of cells that die is established.

• To understand the molecular mechanisms that give rise to programmed cell death (apoptosis)

• To relate dysfunctions in apoptosis with the acquisition of human diseases.

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INTRODUCTION

The cell cycle culminates with cell division; but the cells of a tissue are not in a process of constant cell division (with exception of progenitor cells in tissues with a high cell renovation rate), because the maintenance of **homeostasis** implies an equilibrium between the cells that die and the cells that are generated from cell division. The cells that are lost because of the deterioration caused by senescence, by external tissue damage of any type or by pathological phenomenon (such as infection or mutation) are replaced by cellular division and differentiation through highly regulated processes.

The phenomenon leading to **cell death** can be classified into two large types: a) those that cause the death through external damage that leads to a break in the cell, and are encompassed within what is called necrosis, and b) cell death that is genetically programmed by the cell, called apoptosis. The other type of genetically controlled cell death is mediated by the activity of lysosomes and is called autophagy, whose relevance has been explained in chapter 10.

The concept and physiological importance of apoptosis

As we have indicated at the beginning of the chapter, the equilibrium between the rates of proliferation and cell death depends on the growth and adequate regulation of each organ. This balance requires, in some occasions, an "altruist" behaviour on behalf of the individual cell in favor of the rest of the population, which implies **programmed death** in some cells (in particular if they are potentially dangerous or harmful to the body). In 1972, Kerr and his collaborators introduced the term apoptosis, differentiating it from the unique term used until them to refer to cell death: **necrosis**. Later studies have defined differences between these types of cell death. Necrosis is a type of violent death, quick and accidental, that occurs in response to a severe alteration in the surrounding conditions or due to physical trauma. Death by necrosis takes place in a non-physiological manner; within it, the cells are edematised and liberate their cytoplasmic content towards the extracellular medium, generating the death in part of the tissue, as well as inflammation.

Apoptosis however, is a type of programmed cell death subject to a very strict regulation on behalf of the cell. It is an innate mechanism whose function is to eliminate the cells that are no longer necessary, that have developed in an inappropriate way, or those in which irreparable genetic damage has occurred. It is a controlled process, active and energy dependent, that occurs in a regular way during embryonic development, the remodeling of tissues, the regulation of the immunity response and the progression of tumors. A particularly instructive example of the implications of apoptosis in the development of humans is the formation of independent fingers through the mass death of cells in the interdigital mesenchymal tissue. Other examples are the formation of the brain, during which half of the formed neurons die in advanced stages of development. Given their physiological importance, the understanding of the factors that induce apoptosis is key to understand their implication in the apparition of diseases

As we have commented previously, there is another type of cell death that is controlled by the cell that implies an auto digestion by lysosomes and that is called **autophagy**. This type of death and the mechanisms that carry it out are discussed in chapter 10.

Characteristics of apoptosis

The cells that undergo the process of apoptosis are characterized by the development of particular sequences of morphological and biochemical changes. The most typical morphological features are the formation of ***blebs*** in the plasma membrane and the **condensation of the chromatin** that leads to very electrondense patches **(figure. 1)**. In addition, the cellular volume decreases, and in the final phases of the process, the **apoptotic bodies** form (corresponding to cell remains encapsulated by fragments of the plasma membrane), which are finally phagocytosed by neighboring cells or the immune system, preventing in this way an inflammatory response.

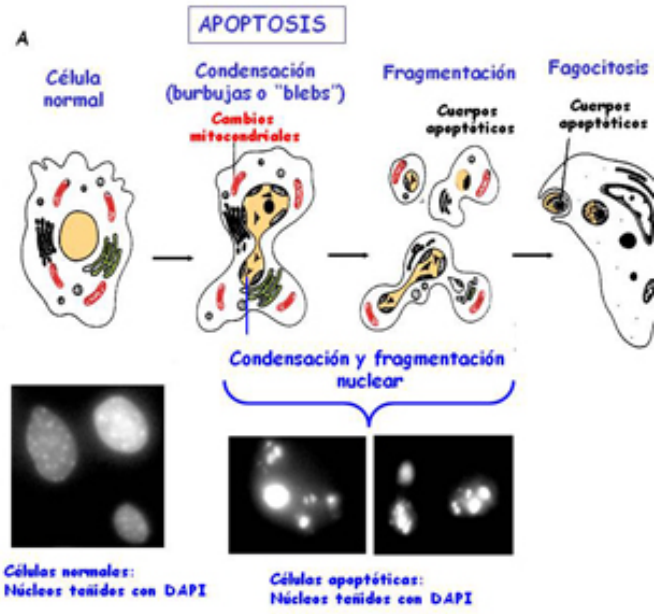


Figure 1. Morphological changes during apoptosis.

To recognize cells in apoptosis through morphological observation we frequently use an electron microscope, as it permits us to easily identify the aforementioned morphological features. Under a light microscope the cells in apoptosis can be identified by the condensation of chromatin (which appear strongly dyed by hematoxylin when hematoxylin-eosin is used) and by nuclear fragmentation. But to determine with greater precision the presence of apoptotic cells under the light microscope we normally use molecular

techniques or immunohistochemistry to identify proteins activated during apoptosis (Figure 2A).

Amongst the **biochemical alterations** of the programmed cell death, it would be important to mention the **change of location of the phosphatidylserine** which is normally found in the internal part of the plasma membrane, which translocated to external face. The reduction in cytoplasmic volume occurs because of a loss of ions (K+, Cl- and organic ions), which leads to the loss of water. The DNA of apoptotic cells undergoes condensation and posterior fragmentation into units with a characteristically high molecular mass (50kb) and others with characteristically low molecular masses (the internucleosomic fragmentation of 200 base pairs). This gives rise to a pattern of bands with a "ladder-like" arrangement when electrophoresis has been applied. The other key biochemical event, responsible for the majority of the alterations mentioned earlier, is the specific **breaking of multiple cellular structures**, carried out by a family of cysteine-proteases called **caspases** (see later on). These proteins are able to disintegrate the cytoskeleton, nuclear lamina, the Golgi apparatus, the inhibitors of DNAses, etc. Furthermore, they eliminate the cell-cell contacts, inhibit DNA replication and cellular survival signals, and activate endonuclease that degrade DNA.

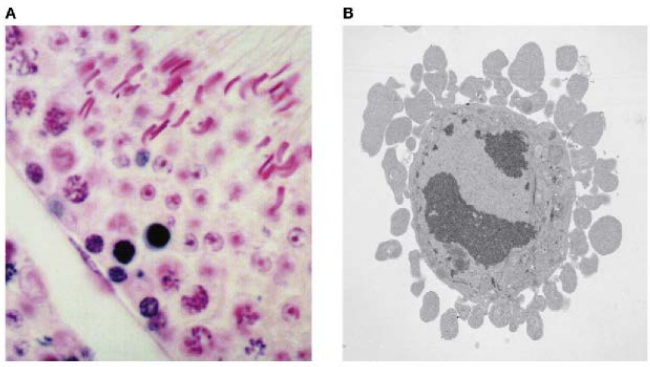


Figure 2. A. Stain of cells in apoptosis (dark coloured) identified by a molecular technique. B. Electron microscope picture of a cell in apoptosis.

The molecular mechanisms of apoptosis

There is a considerably complex chain of events that is in charge of the process of apoptosis. In this overview, we will focus on the most relevant proteins for this first course of Medicine. The triggering of this process depends on the integration of intra- and/or extracellular signals that finally leads to the activation of caspases, which start the proteolytic process. We can distinguish two types of cell pathways for the induction and death by apoptosis: **the extrinsic route and the intrinsic route.**

EXTRINSIC ROUTE

This route is activated by the activity of external molecules. In this way, some cells are induced to die upon **receiving these signals** from neighboring cells or when they are no longer being exposed to the inhibitory factors of apoptosis. Extrinsic signaling is mediated by the binding of specific ligands to **death receptors**, membrane receptors that trigger apoptosis when activated (Figure 3). These death receptors belong to a superfamily of tumor necrosis factor receptors (TNFRs), within which we find **Fas/CD95**, that plays a key role in apoptosis in the **immune**

system. Cell death induced by Fas participates, for example, in the selective elimination of cells of the immune system (cells infected by a virus, tumor cells or lymphocytes after an immune response).

The initiation of intracellular signaling takes place through the cytoplasmic domain of the receptor, that possesses a sequence called the **death domain** (DD). The adaptor molecules, such as **FADD**, possess their own death domain, thanks to which they are recruited by the active receptor, forming in this way the death inducing signaling complex (DISC). This then initiates the activation of effector caspases that will finally lead to cell death by apoptosis.

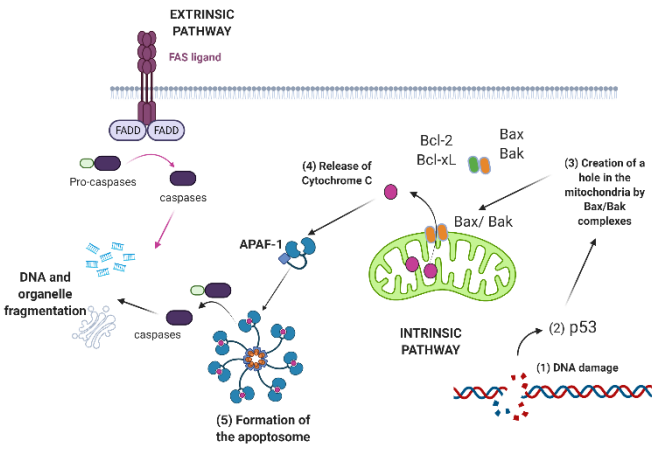


Figure 3. Molecular pathways that lead to apoptosis. The extrinsic pathway relies on external signals. The intrinsic pathway is activated by DNA damage or cellular stress and depends on mitochondrial alteration.

INTRINSIC OR MITOCHONDRIAL ROUTE

The intrinsic route is primarily induced through cell stress, DNA damage and the absence of growth factors in the medium. **DNA damage**, that carries the great risk of possibly transmitting genetic defects to daughter cells through mitosis, is one of the principle inducers of cell stress. An essential protein in the response to DNA damage is **p53**, that can cause the cell cycle to stop if the damage is reparable or can lead to apoptosis if not. p53 therefore participates in the elimination of cells that are genetically anomalous and potentially dangerous. The activation of p53 at the same time gives rise to the activation of **Bax** and **Bad** (proapoptotic proteins) and the inactivation of **Bcl-2** and **Bcl-xL** (antiapoptotic proteins). The translocation of proapoptotic proteins to the mitochondria and their formation into oligomers destabilizes the mitochondria membrane through the formation of **pores** (Figure 3), which results in the freeing of internal mitochondrial proteins such as **cytochrome C**. This joins to the cytoplasmic protein **Apaf-1** (apoptosis protease-activating factor-1), that through an ATP-dependent reaction oligomerizes (forming a multiproteic complex) constituting what is known as the **apoptosome**. Finally, the apoptosome activates caspases resulting in the death of the cell.

Characteristics of some of the molecules that regulate apoptosis

Bcl-2 family

The term **Bcl-2** comes from B-cell lymphoma/Leukemia 2 gene and makes reference to an oncogene that was found associated to a locus of immunoglobulins generated through the translocation t(14:18) in lymphomas of human B cells. Before then, it had been considered that oncogenes belonged to a class of genes that induced cellular proliferation; a characterization of Bcl-2 meant the appearance of a new type of oncogenes with the capacity to prevent cell death. This superfamily, that includes more than 20 members, divides into three groups that

encapsulate both proapoptotic proteins and antiapoptotic proteins.

1. **Antiapoptotic members:** this group includes Bcl-2 and Bcl-XL proteins. The function of the antiapoptotic members is to capture their proapoptotic counterparts, forming heterodimers (e.g., Bcl-2/Bax). From a clinical standpoint, **Bcl-2** has a great value as biomarker. Bcl-2 expression is related with bad prognostic in several types of cancer. Bcl-2 expression also has a predictive value in the absence of a therapeutic response to chemotherapy in some types of tumors. The importance of this protein in cancer has led to the development of specific pharmaceuticals to inhibit Bcl-2.

2. **Proapoptotic members**: They are found as monomers in the cytosol. As previously mentioned, the oligomerization of Bax and Bak opens pores in the external membrane of mitochondria. Cytochrome C is then released to the cytoplasm triggering apoptosis.

Caspases: key molecules for initiation and execution of apoptosis.

The term **caspases** comes from cysteine dependent aspartate-specific proteases and references a family of proteins that, as we have mentioned earlier, possess a special relevance in the process of programmed cell death. It is so important that cell death is only truly characterized as apoptotic if involves caspase activation. Their catalytic activity depends on a cysteine residue located in a highly conserved pentapeptide in the active site of the enzyme. Pro-caspases are the inactive form of the protein. Activation of pro-caspases requires a cleavage that leads to the active protein with pro-apoptotic functions.

Diseases related to alterations in apoptosis

Anomalies in the regulation of apoptosis have been related with the onset of multiple diseases. Pathologies in which there is a reduction of apoptosis and its processes are for example, cancer (where the mitosis/apoptosis ratio is greater than in healthy tissues), and autoimmunity or persistent infections. On the contrary, an excessive and inadequate activation of apoptosis generate neurodegenerative pathologies (Alzheimer, Parkinson, Huntington, lateral amyotrophic sclerosis), autoimmunity, AIDS (exhaustion of T lymphocytes) and ischemia.

The majority of chemotherapeutic pharmaceuticals which are used in the treatment of cancer activate apoptosis in tumoral cells. Resistance to antitumor therapies is normally due to the incapacity to activate proapoptotic routes.

As we have seen in this chapter, apoptosis is a phenomenon of great complexity, with multiple inducing factors, along with others that cause their inhibition. In order to alter this equilibrium and activate apoptosis, an unbalancing between the quantity of pro- and anti-apoptotic factors needs to occur with a slight dominance of the first. A deeper understanding of the mechanisms that affect apoptosis, favoring the appearance of disease, would allow the development of therapeutic strategies to restore this balance.